ITEC 235-940 System Administration

Red Hat Enterprise Linux Security Lab

Managing Services, Ports, and Zones

In this lab, you will first determine the current active zone. Then you will add and activate a permanent rule to allow TCP port 80 HTTP and a runtime rule for traffic intended for TCP port 443 HTTPS. You will add a permanent rule to the internal zone for TCP port range 5901 to 5910. Then you will confirm the changes and display the contents of the affected files. Lastly, you will switch the default zone to the internal zone and activate it.

1. Determine the name of the current default zone by entering the following command: `sudo firewall-cmd --get-default-zone`

2. Add a permanent rule to allow HTTP traffic on the default port, port 80 by entering the following command:
   `sudo firewall-cmd --permanent --add-service http`
   This created a copy of the public.xml file from the /usr/lib/firewalld/zones directory into the /etc/firewalld/zones directory.

3. Activate the rule created in step 2 by entering the following command: `sudo firewall-cmd --reload`

4. Confirm the activation of the new rule by entering the following command: `sudo firewall-cmd --list-services`

5. Display the content of the default zone file to confirm the addition of the permanent rule by entering the following command:
   `sudo cat /etc/firewalld/zones/public.xml`

6. Use the following command to add a runtime rule to allow traffic on TCP port 443: `sudo firewall-cmd --add-port 443/tcp`

7. Verify the runtime rule by entering the following command: `sudo firewall-cmd --list-ports`

8. Add a permanent rule to the internal zone for TCP port range 5901 to 5910 by entering the following command:
   `sudo firewall-cmd --add-port 5901-910/tcp --permanent --zone internal`

9. Display the content of the internal zone file to confirm the addition of the permanent rule by entering the following command:
   `sudo cat /etc/firewalld/zones/internal.xml`

10. The firewall-cmd command creates a backup of the modified zone file using a .old extension. Enter the following command to switch the default zone to internal: `sudo firewall-cmd --set-default-zone internal`

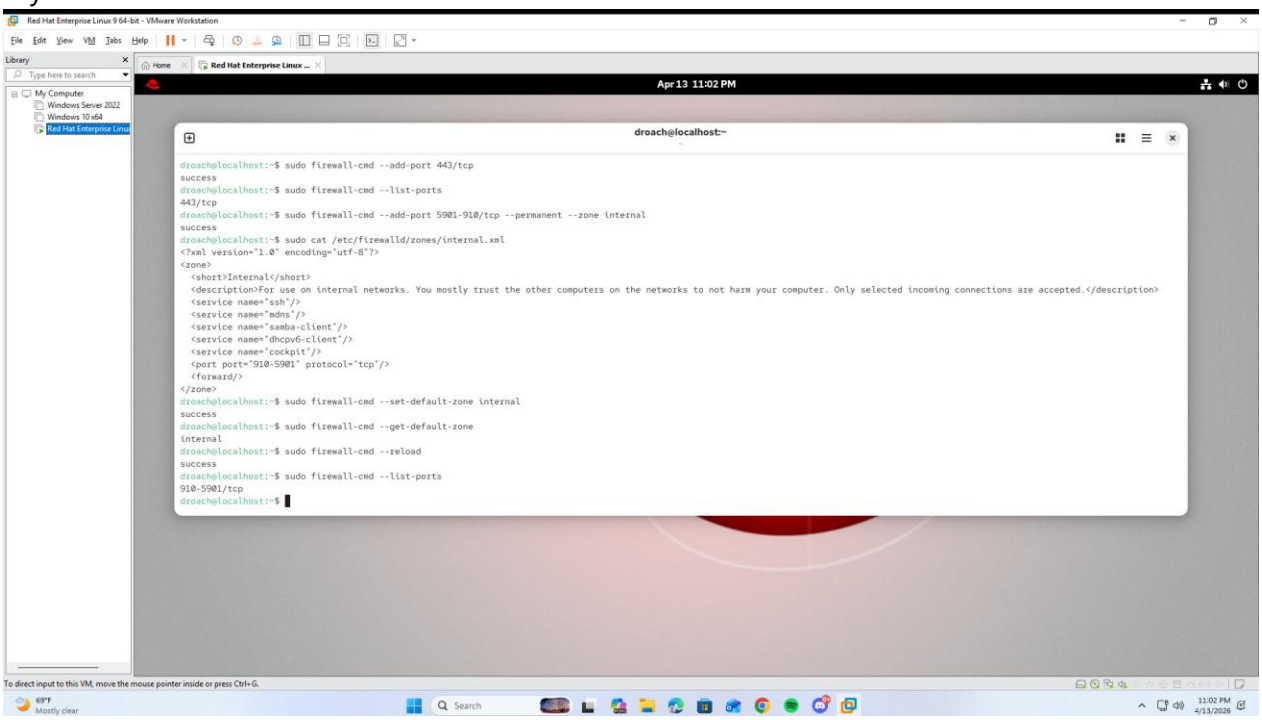
11. Enter the following command to verify that zones have been changed: `sudo firewall-cmd --get-default-zone`

12. Activate the rules defined in the internal zone and list the port range added in step 8 by entering the following command:
    `sudo firewall-cmd --reload`

13. Enter the following command to verify the ports. `sudo firewall-cmd --list-ports`

14. **<u>Create a screen capture of the port list and submit it using the lab submission link.</u>**

My Answer:



15. Remove the permanent rule for HTTP from the public zone by entering the following command: `sudo firewall-cmd --remove-service=http --zone public --permanent`

16. Remove the permanent rule for ports 5901 to 5910 from the internal zone by entering the following command:
    `sudo firewall-cmd --remove-port 5901-5910/tcp --permanent`

17. Enter the following command to switch to switch the default zone to public: `sudo firewall-cmd --set-defaul-zone=public`

18. Verify the public zone is the default zone by entering the following command: `sudo firewall-cmd --get-default-zone`

19. Enter the following command to activate the public zone rules: `sudo firewall-cmd --reload`

20. Enter the following command to view the list of services: `sudo firewall-cmd --list-services`
    Notice that the http service has been removed.

21. **Create a screen capture of the service list and submit it using the lab submission link.**

My Answers:

